Trainings

General Information

with Mechanically Actuated Injector

The barring mechanism is inside the front cover. The front cover **must** be removed from the engine to service or rebuild the barring mechanism.

This barring mechanism contains a spring loaded worm gear. The worm gear engages the camshaft gear when the barring shaft is pushed in the front cover and turned in a **counterclockwise** direction. The barring mechanism will **only** turn the engine in the direction of normal rotation. Turn the barring shaft in a **clockwise** direction to disengage the worm gear. If the worm gear remains engaged accidently during the engine start, the engine rotation will disengage the parts without damage.

The static injection timing is adjusted by using different camshaft keys. The selection of a key will change the position of the camshaft lobes in relation to the timing mark on the camshaft gear. The gear **must** be removed to change the injection timing.

The camshaft end clearance is determined by the clearance between the camshaft and the thrust plate. The camshaft gear **must** be removed to adjust the camshaft end clearance.

Note : The camshaft does not have to be removed to remove the camshaft gear. Use the camshaft gear puller, Part Number 3376400.

Camshafts that are damaged or worn on the injector or the valve lobes **must** be replaced. Cummins® Inc. does **not** recommend the grinding of camshaft lobes.

The crankshaft uses forged counterweights.

Oversize main bearings and thrust bearings are available for service. Cummins® Inc. recommends regrinding all of the main or the connecting rod journals when one requires regrinding.

The vibration damper controls the twisting or torsional vibration of the crankshaft. A vibration damper is engineered for use on a specific engine model.

It is **not** economical to repair a vibration damper in the field. Install a new or a rebuilt vibration damper if the inspection indicates that a damper is defective.

The viscous vibration damper has a limited service life. The damper **must** be replaced.

with Electronically Actuated Injector

The present QSK19 cylinder block has been modified to include an engine speed sensor bore at the left rear of the engine that is just below and in back on the lube oil filter head.

A modified QSK19 crankshaft has been incorporated into the new engine configuration. The new design includes a tone wheel at the rear of the crankshaft for the engine speed sensor. The tone wheel is **not** serviceable in chassis.

A modified K19 camshaft has been incorporated into the new engine configuration. The camshaft has been modified by removing the injector lobes, which are no longer needed with the electronic injectors. These injector lobes have now been machined to a circular pattern for piloting the camshaft during engine assembly. A single-piece camshaft cover continues to be used for this engine configuration.

A modified K19 Series cam follower assembly has been incorporated into the new engine configuration. The center injector follower has been removed and a spacer has been added since the injectors are no longer mechanically actuated.

Single piece ferrous cast ductile iron pistons have been incorporated into the new engine configuration. Ferrous cast ductile iron pistons are also used on all other high horsepower engine products.

A modified K19 Series cylinder head has been incorporated into the new engine configuration. The cylinder head has been modified to accommodate the new injectors, valve guides, valve stem seals, and fuel drain lines. This revised K19 cylinder head requires three less drillings. Two were used for mounting the fuel rail and the third was required for the fuel drain. The single fuel drilling that was retained has been machined to accept banjo fittings that are utilized by the fuel drain. Additionally, cast iron valve guides and new stem seals are used to improve the valve durability as well as emissions. The cylinder head also has a metric threaded hole for the injector hold-down. The injector hold-down capscrews have also changed.

The rocker housing has been modified to facilitate single cylinder repair without having to remove adjacent rocker housings. It also uses a four bolt mounting arrangement for improved sealing characteristics.

The K19 Series rocker levers have been incorporated into the new engine configuration; however, the push rods and rocker levers associated with the injection cycle have been removed. The injector rocker levers have been replaced by spacers.

A new cast aluminum two-piece valve cover has been designed to accommodate the new injectors. Additionally, the breather arrangement has been cast into the valve cover, which will employ a metallic mesh as the filtration media. The Original Equipment Manufacturer (OEM) **must** connect a 25.4 mm [1 in] inside diameter (ID) hose (capable to 149° C [300 °F]) to each of the three breather ports. The valve cover is now attached to the rocker housing using a four bolt clamping pattern.

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